

Standard Normal

Grinnell College

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Review

The **Law of Large Numbers** guarantees that, as the number of observations n in my sample increases, my estimate of the sample mean will converge to the true mean

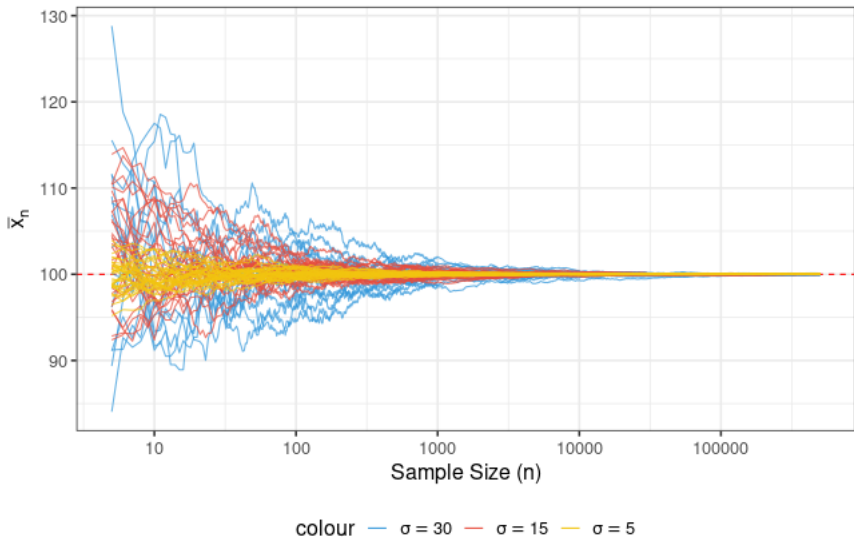
A **sampling distribution** refers to the distribution of a sample statistic (i.e., $\bar{X}$) if we were to repeatedly sample from a population and recompute the statistic

- ▶ What values would they take?
- ▶ How frequently would they appear?

The **Central Limit Theorem** states that if my statistic is an average or a proportion, then the sampling distribution of my statistic will be approximately normal, with

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

Different Sample SD



Two Types of Questions

Really, we have two separate questions:

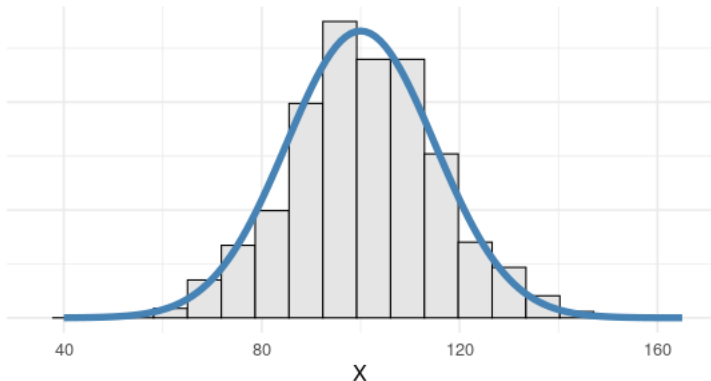
1. For a sample of size n , what happens to the uncertainty in my estimate of μ as $n \rightarrow \infty$
2. If instead n is fixed and I repeatedly collect many samples of the same size, what will the distribution of my estimate of μ look like?

Why are these things I'm interested in knowing?

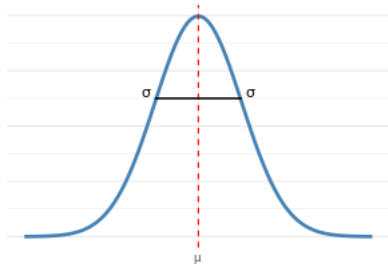
Notes on Normal

The **normal distribution** describes a distribution that is

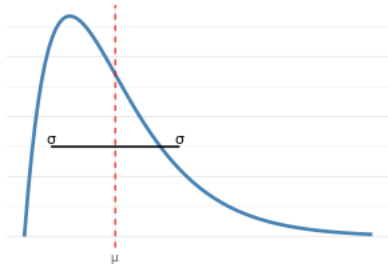
- ▶ Bell-shaped
- ▶ Symmetric about the mean
- ▶ Has two distributional parameters, the mean μ and standard deviation σ



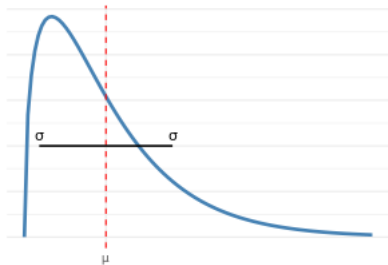
Normal



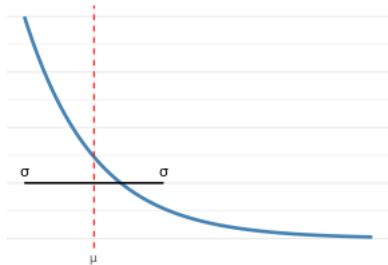
Gamma



Chi-Square



Exponential



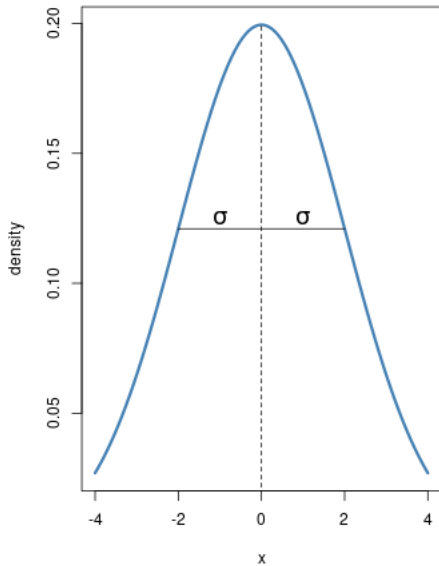
Some Terms to Know

Standard Deviation: A description of the variability in our *observations* describing average distances from the average or mean. It is often denoted σ

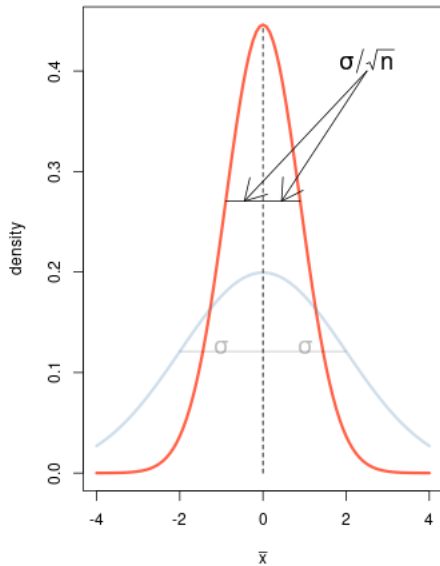
Standard Error: A description of spread in our *sampling distribution*. We will denote standard error as SE , with $SE = \sigma/\sqrt{n}$, where n is the number of observations in our sample.

Note that the standard error *is* the standard deviation of the sampling distribution

Standard Deviation



Standard Error



Standardization

As a consequence of the fact that, *asymptotically*,

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

I know that if I were to collect many many samples of $\bar{X}$, the average value of these $\bar{X}$ would be near μ . Further, the “typical distance” of each $\bar{X}$ from μ would be $\sigma/\sqrt{n}$

Suppose each of us collect our own sample mean of the same size. What would happen if we standardized these values, taking

$$Z_i = \frac{\bar{X}_i - \mu}{\sigma/\sqrt{n}}?$$

Standard Normal Distribution

The **standard normal distribution** (typically represented with the variable Z) is normal distribution with a mean value of 0 and a standard deviation of 1

$$Z \sim N(0, 1)$$

Because standardizing a distribution only shifts and scales the distribution (without changing it's shape), it follows that

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Let's be clear

To be absolutely clear, it's not standardizing our sample mean that makes it normal, it's the CLT. The logic goes:

1. **CLT:** For “sufficiently large n ” the distribution of $\bar{X}$ is approximately normal with mean μ and standard deviation (standard error) $\sigma/\sqrt{n}$
2. **Standardization:** Standardizing *any* variable will create a new variable Z that has a mean value of 0 and a standard deviation of 1 without changing its shape
3. Therefore, since $\bar{X}$ is approximately normal for sufficient n , it follows that standardizing $\bar{X}$ will give us:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Example – Penguin

Suppose I have a population of Gentoo penguins with a true average bill length of $\mu = 47.56$ millimeters and a standard deviation of $\sigma = 3.1$ millimeters

Suppose I collect a sample of size $n = 20$ and find my sample mean of $\bar{x} = 48.94$. Is this a value I should expect to get from a random sample? Or does it appear to be an outlier?

Example – Penguin

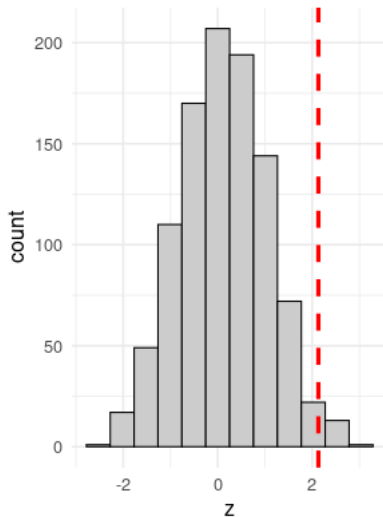
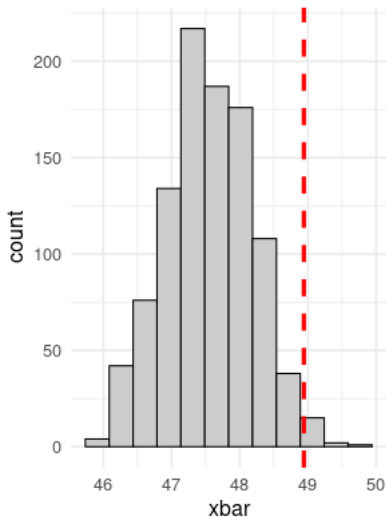
We can start by standardizing to see how this sample mean falls relative to the distribution of means we might get

$$\begin{aligned} Z &= \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{48.94 - 47.56}{3.1 / \sqrt{20}} \\ &= 1.99 \end{aligned}$$

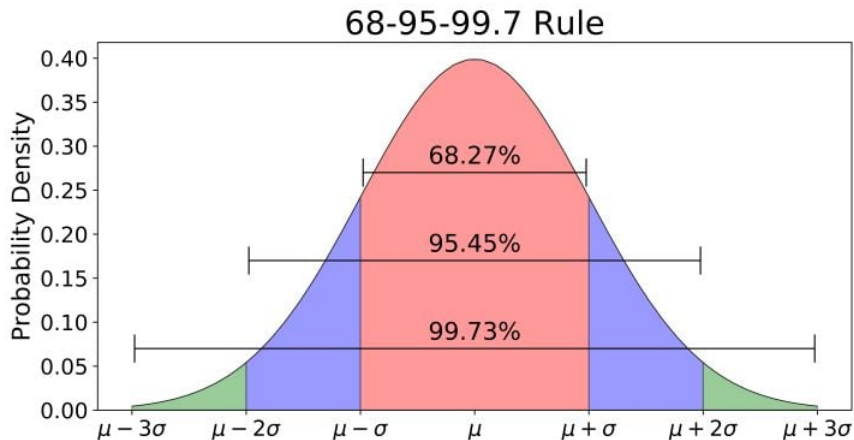
Indicating that the mean of this sample is about 1.99 standard deviations away from what we would expect

Example – Penguins

Here, I collected 1,000 samples of $n = 20$ penguins and found the mean of each



Empirical Rule



A Simulation of Sorts

We see from a simulation of sampling, from the 1000 standardized values we collected, 693 were within one standard deviation and 953 were within 2 standard deviations

```
1 > xbar <- replicate(1000, {
2 +   sample(gentoo$bill_length_mm, size = 20, replace = TRUE)
3 +   %>% mean()
4 + })
5 > z <- (xbar - 47.46) / (3.1 / sqrt(20)) # Create std. values
6 >
7 > sum(z < 1 & z > -1)
8 [1] 693
9 > sum(z < 2 & z > -2)
10 [1] 953
11 > length(z)
12 [1] 1000
```

Unlikely values

Suppose I collect a sample and find that the standardized sample mean is $Z = 5$

- ▶ What does this mean?
- ▶ Is this possible?
- ▶ Does it mean something is wrong with my sample?
- ▶ How to decide?

AHHHHHHHHHH!!!!!!!!!!!!!!!!!!!!!!